MILITARY SPECIFICATION

POWER SUPPLY ASSEMBLY (6A2)

T4138 TEST SET

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of Power Supply Assembly (6A2) which is part of the T4138 Test Set.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

DRAWINGS

ORDNANCE CORPS

8833322	- 6.0, 7.0 Volt Regulator Assembly.
8833323	- 350 V, -10 V, 28 V and 56 V Regulator Assembly.
8833324	- 28 V Output Assembly.
8833325	- 400 Cycle Inverter Assembly
8833326	- Pack and Panel Assembly.
8847624	- Rotary Vacuum Pump.
8848006	- 95 Volt Input Regulator Assembly.
8848035	- 1000 Cycle Inverter No. 1 Assembly.

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8848055 - 1000 Cycle Inverter No. 2 Assembly.
8848094 - 330-380 Volt Regulator Assembly.
8849206 - Thermostat.
8849230 - Solenoid Valve Type V51.
8849250 - Reset Switch.
8850405 - Rack and Panel Assembly.
EL-8854168 - Equipment List.
8858809 - Rack and Panel Assembly.
8857106 - Probe Adapter Assembly.
8883498 - 95 Volt Input Regulator Assembly 6A2A1.
8883499 - 1000 Cycle Inverter # 1 & 2 Assembly 6A2A3.
8883500 - 330-380 Volt Regulator Assembly 6A2A4.
8883501 - 6.0, 7.0 Volt Regulator Assembly 6A2A5.
8883502 - 350 V, -10 V, -28 V and 56 V Regulator Assembly 6A2A6.
8883503 - 28 V Output Assembly 6A2A7.
8883504 - 400 Cycle Inverter Assembly 6A2A8.
8883559 - Power Supply Assembly 6A2.
8886903 - Fan.

PUBLICATIONS

ORDNANCE CORPS

TB ORD 1026/121 - Calibration Procedure.

(Copies of specifications, standards, drawings, and publication required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. Each of the following listed assemblies shall comply with all requirements specified in the tabulated drawings, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein:

No.	Assembly	Drawing
6A2	Power Supply Assembly	8847714
6A2A1	95 V Input Regulator Assembly	8848006
6A2A2	1000 Cycle Inverter No. 1 Assembly	8848035
6A2A3	1000 Cycle Inverter No. 2 Assembly	8848055
6A2A4	330-380 V Regulator Assembly	8848094
6A2A5	6.0, 7.0 V Regulator Assembly	8833322

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6A2A6	350V,-10V,-28V and 56V Regulator Assembly	8833323
6A2A7	28 V Output Assembly	8833324
6A2A8	400 Cycle Inverter Assembly	8833325
6A2A9	Rack and Panel Assembly	8833326

3.3 First article testing. Requirements for submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Operational characteristics.

3.5.1 Module assemblies. Each of the assemblies listed in 3.2 above shall pass the operational characteristics as defined in 3.5 of the appropriate appendix as listed below.

No.	Assembly	Appendix
6A2	Power Supply Assembly	A
6A2A1	95 V Input Regulator Assembly	B
6A2A2	1000 Cycle Inverter No. 1 Assembly	C
6A2A3	1000 Cycle Inverter No. 2 Assembly	D
6A2A4	330-380 V Regulator Assembly	E
6A2A5	6.0, 7.0 V Regulator Assembly	F
6A2A6	350, -10, -28 and 56 V Regulator Assembly	G
6A2A7	28 V Output Assembly	H
6A2A8	400 Cycle Inverter Assembly	I
6A2A9	Rack and Panel Assembly	J

3.5.2 Spare module assemblies. Modules designated as spare parts shall meet the requirements of 3.2 and 3.5.1. In addition, each spare module must be tested for compatibility with and in a T4138 test set. The compatibility test shall be conducted in accordance with 4.3.3.2.2.

3.6 Storage transit characteristics. Each assembly or power supply shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The assemblies shall also comply with 3.5 before and after each environmental test. Testing shall be as specified in 4.4.3.

3.6.1 High temperature. Each assembly shall be capable of withstanding a storage temperature of 160 ± 5 degrees Fahrenheit (°F.) without damage and shall comply with 3.5 of the appropriate appendix after having been subjected to this temperature and then operated at 130 ± 5°F. The test shall be conducted as specified in 4.4.3.1.

3.6.2 Low temperature. Each assembly shall be capable of withstanding a storage temperature of -65 ± 5°F. without damage and shall comply with 3.5 of the appropriate appendix after having been subjected to this temperature and

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then operated at 0 + 5°F. The test shall be conducted as specified in 4.4.3.2.

3.6.3 Vibration. Each Power Supply Assembly 6A2 (or module if procured as spare) in its shipping container (see 5.1) shall be capable of withstanding vibrations as specified in Table I without evidence of damage and shall comply with 3.5 of the appropriate appendix after vibration. The vibration shall be applied in the vertical axis as specified in 4.4.3.3.

Table I

Hertz	Sweep Time (Minutes)	Double Amplitude (Inches)	Number of Sweeps
5-10-5	10 + 0.5	0.4 + 10%	1
11-15-11	10 + 0.5	0.25 + 10%	1
16-20-16	10 + 0.5	0.13 + 10%	1
21-60-21	15 + 0.5	0.06 + 10%	1

3.6.3.1 Cycling. Each assembly shall be capable of withstanding cycling vibration through a frequency range of 8-55-8 Hertz (Hz) at a double amplitude of 0.060 + 0.006 inch and 55-500-55 Hz at 3.0 + 0.5 gravity units (g's) for a period of 15 minutes minimum in each of three (3) mutually perpendicular axes. The vibration shall be performed in accordance with 4.4.3.3.1. After the vibration, the assembly shall comply with 3.5 of the appropriate appendix and shall show no evidence of damage.

3.6.3.2 Vibration at resonance. Each assembly shall be capable of withstanding resonant vibration at a double amplitude or acceleration for three mutually perpendicular axes. The test shall be performed as specified in 4.4.3.3.2. After the vibration, no damage to the assembly shall be evident and it shall comply with 3.5 of the appropriate appendix.

3.6.4 Transportation shock. Each assembly shall be capable of withstanding three shocks of 30 + 5 g's in both directions in each of three mutually perpendicular axes for a total of 18 shocks. The time duration of each shock shall be 11 + 2 milliseconds (ms) as measured between points which are ten percent minimum of peak value. The time to peak shall be 5.5 + 1 ms. The test shall be performed as specified in 4.4.3.4. After the test, the assembly shall show no evidence of damage and shall comply with 3.5 of the appropriate appendix.

3.6.5 Humidity. The 6A2 Power Supply Assembly shall withstand exposure to humidity without evidence of deterioration and shall comply with 3.5 of Appendix A when tested as specified in 4.4.3.5.

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3.7 Soldering. All soldering shall comply with MIL-STD-1460.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.8.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 2 complete assemblies (see dwgs listed in 3.2) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II.

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Table II

First article inspections and tests

Inspection or Test	Dest	Non Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Operational characteristics (see 3.5)		X
3. High temperature (see 3.6.1)	X	
4. Low temperature (see 3.6.2)	X	
5. Vibration (see 3.6.3)	X	
6. Transportation shock (see 3.6.4)	X	
7. Humidity (see 3.6.5)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as in 4.3.2.1 of the appropriate appendix.

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Calibration capability	Major
b. Operational limits	Major

4.3.3.1 Calibration capability. Each assembly, unless specified otherwise, shall be calibrated in accordance with 4.4.1 of the appropriate appendix. If any item fails to comply with 3.5.1 of the appropriate appendix, the item shall be rejected and removed from the lot.

4.3.3.2 Operational limits.

4.3.3.2.1 Module assemblies. Each assembly shall be tested in accordance with 4.4.2 of the appropriate appendix. If any item fails to comply with 3.5.2 of the appropriate appendix, the item shall be rejected and removed from the lot.

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4.3.3.2.2 Spare module assemblies. Connect the test set to the test simulator for T 4138 (see Figure 1). Install the module to be tested in a T 4138 Test Set that is capable of meeting the requirements of MIL-P- (MU). Operate the test set through one complete missile test with no-failure parameters inserted. If a green "GO" light is obtained, the module meets the compatibility requirement. If a green "GO" light is not obtained, the module shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8854168). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Calibration capability.. The procedures for determining calibration capability of each assembly are shown in the appropriate appendix.

4.4.2 Operational limits. The procedures used in testing for operational limits of each assembly are shown in the appropriate appendix.

4.4.3 Environmental testing (see 3.6).

4.4.3.1 High temperature. Each assembly shall be placed in a test chamber. The temperature of the chamber shall be raised to 160 ± 5°F. and held for a maximum of four (4) hours or until the temperature stabilizes (see 6.4). The temperature shall then be lowered to 130 ± 5°F and held at this temperature for a maximum of four (4) hours or until the temperature has stabilized. At the end of the soak period and while still at 130 ± 5°F, the assembly shall comply with 3.5 of the appropriate appendix.

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4.4.3.2 Low temperature. Each assembly shall be placed in a test chamber. The temperature of the chamber shall be lowered to -65 ± 5°F. and held for a maximum of four (4) hours or until the temperature stabilizes (see 6.4). The temperature shall then be raised to 0 ± 5°F and held for a maximum of four (4) hours or until the temperature stabilizes. At the end of the soak period and while still at 0 ± 5°F, the assembly shall comply with 3.5 of the appropriate appendix.

4.4.3.3 Vibration. Place the assembly in its shipping container on a suitable fixture and vibrate in the vertical axis according to the schedule in Table I. At the conclusion of the test the assembly shall comply with 3.5 of the appropriate appendix.

4.4.3.3.1 Cycling. Each assembly shall be fastened securely at its normal mounting location on a suitable vibration platform such as to permit vibration on the three major axes. A frequency cycling test shall be conducted in which the frequency is varied uniformly through the ranges of 8-55-8 Hz at a double amplitude of 0.060 ± 0.006 inch and 55-500-55 Hz at 3.0 ± 0.5 g's for a period of 15 minutes minimum in each of three mutually perpendicular axes. At the conclusion of the test the assembly shall comply with 3.5 of the appropriate appendix.

4.4.3.3.2 Vibration at resonance. Each assembly shall be fastened securely at its normal mounting location on a suitable vibration platform such as to permit vibration on the three major axes. The resonant frequencies of the assembly shall be determined by varying the vibration frequency for 8-55-8 Hz at a double amplitude of 0.060 ± 0.006 inch and 55-500-55 Hz at 3.0 ± 0.5 g's in each of the three perpendicular axes. The assembly shall withstand resonant vibration for 3 minutes minimum in each of the three mutually perpendicular axes at the frequencies and amplitudes previously determined to be the most severe for each axis. If no resonance is found, the assembly shall be vibrated for 3 minutes minimum in each axis at 55 Hz and 500 Hz for a total of 9 minutes. Half of the total vibration time will be at 55 Hz and half at 500 Hz. At the conclusion of this test, the assembly shall comply with 3.5 of the appropriated appendix.

4.4.3.4 Transportation shock. Each assembly shall be subjected to three shocks of 30 ± 5 g's in each direction of each of three mutually perpendicular axes for a total of 18 shocks. The duration of shock shall be 11 ± 2 ms as measured between points which are 10 percent minimum of the peaks. The time to peak shall be 5.5 ± 1 ms. Following the test, the assembly shall comply with 3.5 of the appropriate appendix. The assembly shall be examined for damage resulting from the shock test.

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4.4.3.5 Humidity. The 6A2 power supply with the cover removed shall be placed in a humidity test chamber and subjected to a 72 hour humidity test. The temperature in the chamber is to be maintained a 80° ± 20°F and the relative humidity at 95 percent plus 4 minus 5 percent. The water used in the control of the relative humidity shall be distilled having a pH value of between 6.5 and 7.5 at 77°F. The pH value shall be adjusted by the use of chemically pure hydrochloric acid or chemically pure sodium hydroxide. After 72 hours min the assembly shall be removed from the chamber. Moisture may be removed by blowing dry filtered compressed air over the unit for 5 minutes max. There shall be no evidence of deterioration and the unit shall comply with the requirements of 3.5.2 of Appendix A.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The assembly shall be packaged in accordance with dwg the applicable drawing listed below using the specified moisture barrier.

5.1.2 Level C. The assembly shall be packaged in accordance with dwg the applicable drawing listed below except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The assembly shall be packed in accordance with dwg the applicable drawing listed below.

5.3 Marking. The 6A2 power supply assembly shall be marked in accordance with dwg 8847714.

No.	Assembly	Packaging Drawing
6A2	Power Supply Assembly	8883559
6A2A1	95 Volt Input Regulator Assembly	8883498
6A2A2	1000 Cycle Inverter #1 Assembly	8883499
6A2A3	1000 Cycle Inverter #2 Assembly	8883499
6A2A4	330-380 Volt Regulator Assembly	8883500
6A2A5	6.0, 7.0 Volt Regulator Assembly	8883501
6A2A6	350 V, -10 V, -28 V and 56 V Regulator Assembly	8883502
6A2A7	28 V Output Assembly	8883503
6A2A8	400 Cycle Inverter Assembly	8883504
6A2A9	Rack and Panel Assembly	Not Applicable*

*Use packing drawing 8858809 for Rack & Panel, P/N 8850405

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-2079 dated 14 September 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A075

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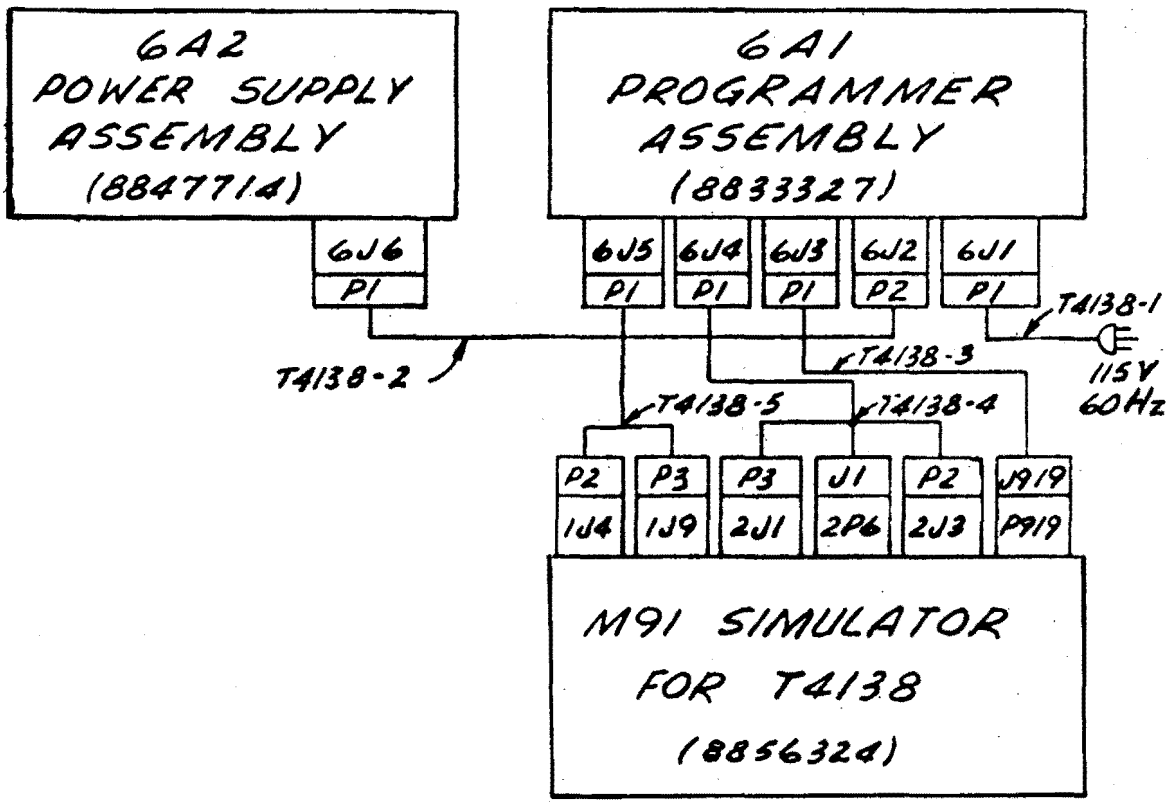


Figure 1

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APPENDIX A

Power Supply Assembly of T4138 Test Set

(6A2)

The following sections apply specifically to the Power Supply Assembly (6A2) which is composed of the modules, described in Appendices B through I, and auxillary equipment, specified in Appendix J. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-P- (MU).

3.5 Operational characteristics.

3.5.1 Calibration capability. The 6A2 assembly shall be capable of calibration using Unit B of Load Box, Electrical: T4138 and Calibration Procedure TB ORD 1026/121. Calibration shall be performed as specified in 4.4.1.

3.5.2 Operational limits. The 6A2 assembly output pins shall be loaded (with resistive loads) through the ranges specified for the individual modules that supply the appropriate 6A2 connector pins (see Figure 1, this appendix). Voltage limits shall remain within the limits specified in 3.5.2 of Appendices E, F, G, H and I with the exception of the maximum allowable ripple voltages which shall not exceed the following values:

6A2 Assembly Terminals	Module Terminals	Max. Ripple
A - w	6A2A4 P-R	500 mv Peak
f-x-y & G-H-h	6A2A5 MM-NN-LL & JJ-FF-HH	100 mv Peak
C - e	6A2A6 CC-DD	360 mv rms
AA-z-j	6A2A6 Y-Z-V	15 mv rms
g - F	6A2A6 U-R	500 mv rms
i - L	6A2A7 CC-DD	20 mv rms
MM-NN & u-GG	6A2A7 X-W & R-P	1.0 mv rms

The 6A2 output loads must be programmed so as to draw not more than the maximum nor less than the minimum power possible, under operational conditions, during each output measurement. Auxillary equipment must function as follows:

Vacuum Pump - Will operate when connecting FF to MM and NN through external switch 1 (see Figure 1, this appendix).

Solenoid Valve - Will operate when connecting DD to MM and NN through external switch 2 (see Figure 1, this appendix).

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Pressure Switch	- When pressure switch makes, after operation of vacuum pump, 56 volts dc will be made available at pins EE and F (see Figure 1, this appendix).
Overtemp feature	- At ambient temperatures that exceed the maximum allowable operating temperature or in the event insufficient cooling air is provided, the input power to the 6A2A1 shall be automatically removed and simultaneously the Overtemp light shall light. Operation shall be restored and the Overtemp light shall go out when the unit has cooled and the Temp Reset button is depressed. Testing shall be performed as specified in 4.4.2.

4.3.2.1. Classification of characteristics.
4.3.2.1.1 Power Supply Assembly (6A2) (see dwg 8847714).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect location of components	Visual
	102. Missing component	Visual
	103. Incorrect location and orientation of connector	Visual
	104. Defective latches	Visual
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual
	202. Component identification markings illegible or obliterated	Visual
	203. Improper finish or color	Visual

4.3.2.1.2 Preparation for delivery (see dwg 8883559).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:

101. Packing material and packaging not proper or missing.....Visual

102. Marking not in accordance with dwg.....Visual

Minor:

201. Evidence of poor workmanship (see 3.8).....Visual

4.4 Test Methods.

4.4.1 Calibration capability. Prior to operational limits testing as specified in this Appendix, the 6A2 Assembly shall be calibrated using Unit B of Load Box Electical: T4165 and Calibration Procedure TB ORD 1026/121.

4.4.2 Operational Limits. For all the 6A2 outputs shown in Figure 1, this appendix, make the input and output adjustments prescribed in 3.5.2, this appendix. Voltage, current, ripple and reponse will be measured as indicated in 4.4.2 of Appendices E, F, G, H and I, MIL-P (MU). Input and output voltages only shall be measured as in 4.4.2 of Appendix B.

4.4.2.1 Overtemp feature. As an operational check the cooling air may be removed thus allowing the thermostat (8849206) contacts in the 6A2A1 module to "make" removing input power and simultaneously lighting the Overtemp light. Cooling air may be removed by removing the blower motor fuse F13. During the operational check the temperature of the edge of the cooler plate in the 6A2A1 module upon which the thermostat is mounted shall be monitored at a position directly above the thermostat using a temperature potentiometer, Brown Instrument type 2736 or equivalent. At no time during the operational check should the plate be allowed to exceed 195°F. before the Overtemp light operates and input power is removed.

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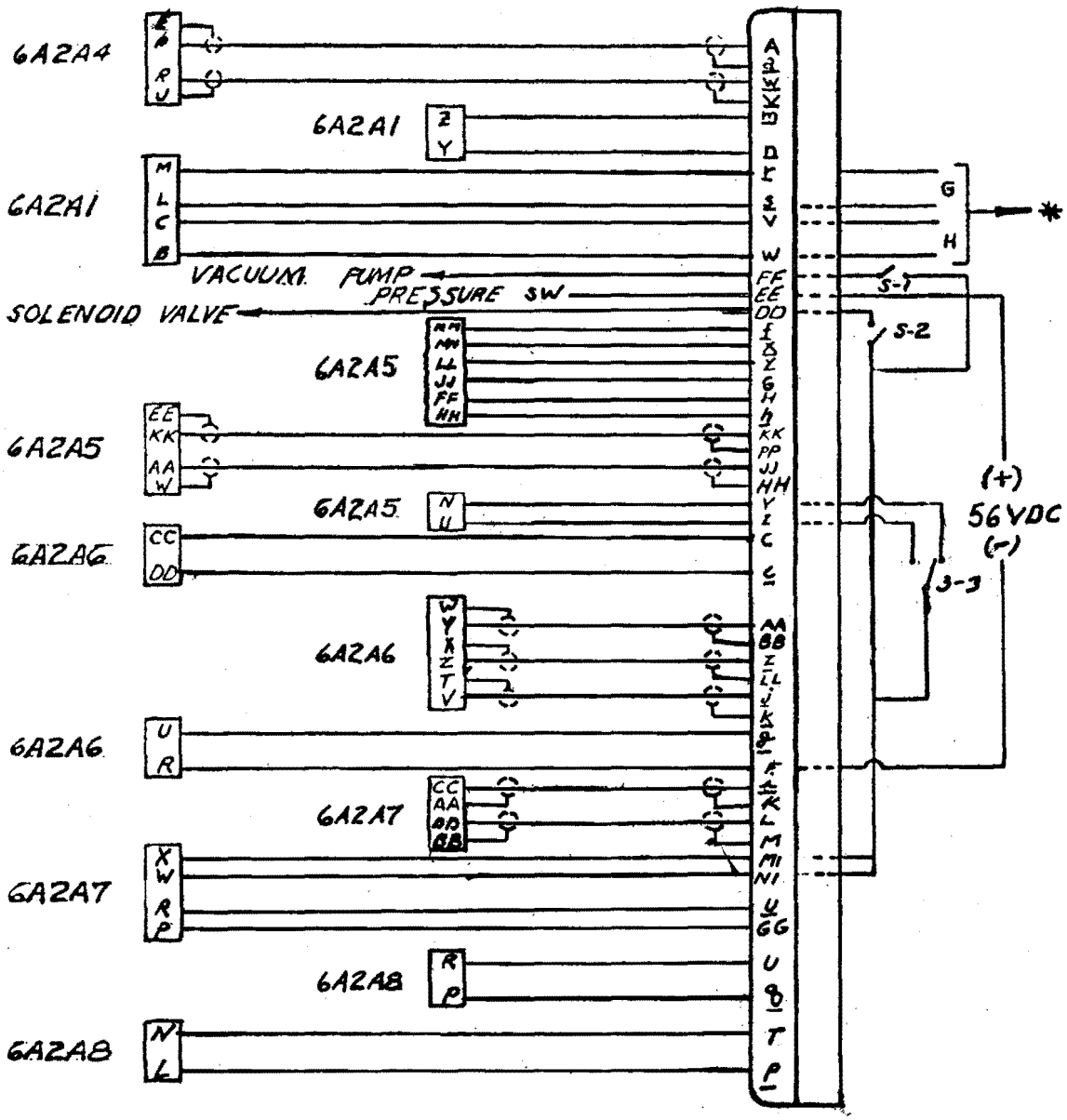


Figure 1 - Suggested test circuit for 6A2 power supply ass'y

* INPUT-115±10VAC 60 HERTZ VARIABLE SOURCE

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APPENDIX B

95 Volt Input Regulator Assembly

(6A2A1)

The following sections apply specifically to the 95 Volt Input Regulator Assembly module. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-P- (MU).

3.5 Operational Characteristics.

3.3.1 Calibration Capability. Under all conditions indicated in 3.5.2 below, operation of calibration pot R12 shall provide output voltage adjustment greater than 96 volts dc and less than 94 volts dc. For calibration, R12 is to be adjusted to provide 95 volts at 5 amps load current. The calibration shall be performed in accordance with 4.4.1.

3.5.2 Operational Limits. With the 6A2A1 Regulator Assembly connected as shown in Figure 1 of this appendix and with the input voltage and load current varied as indicated, the following circuit limits will be maintained. This test shall be performed in accordance with 4.4.2.

Terminals

L, M & B, C	Input Voltage Variation	115 ± 10 vac, 60 Hertz, single phase
Y, AA, CC (Neg)	Output Current Variation	2-8 amps dc
BB, DD, Z	Output Voltage	95 ± 1.9 volts dc
	Ripple Voltage	1.0 volt rms (max.)
	Regulation	± 2%
	Response	Output voltage to be within regulation limits within 1.0 millisecond after disturbance.

4.3.2.1 Classification of characteristics.

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4.3.2.1.01 95 Volt input Regulator Assembly (6A2A1) (see dwg. 8848006).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect location and orientation of connector	Gage
	102. Loose or missing fasteners	Visual/Manual
	103. Defective solder joints	Visual/Manual
	104. Component missing	Visual
	105. Cracked or broken connectors or damaged pins	Visual
	106. Board mounting screws loose	Manual
	107. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual
	202. Component identification markings illegible or obliterated	Visual
	203. Improper finish or color	Visual

4.3.2.1.2 Preparation for delivery (see dwg 8883498).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Packing material and packaging not proper or missing	Visual
	102. Marking not in accordance with dwg	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.

4.4.1 Calibration capability. Connect module as shown in Figure 1 of this appendix and make the adjustments specified in paragraph 3.5.1, this appendix. The module may be manually adjusted for the nominal output voltage by removing the button (8855791) from the module cover (8857395), loosening the calibration pot (R12) lock nut with a 5/16" spin

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tight socket wrench or equivalent and by turning the pot shaft clockwise to increase the output voltage. Ac input voltage readings shall be made with a Hewlett-Packard model 400H, rms, vtvm or equivalent. Dc voltages shall be measured with a Weston, model 931, dc voltmeter or equivalent. Dc to be measured with a Weston, model 45 dc ammeter (with 10 amp shunt) or equivalent. All instrumentation to be isolated from power line ground. The assembly shall be calibrated at a temperature of 80 + 30°F. Ripple voltage to be measured with a Fluke model 910A,rms voltmeter or equivalent.

4.4.2 Operational limits. Connect module as indicated in Figure 1 of this appendix and make the adjustments specified in 3.5.2, this appendix. Voltage and current readings will be measured with the instrumentation described in 4.4.1, this appendix. Transient response is to be measured with a Hewlett-Packard, model 130B oscilloscope and recorded by a Polaroid Land camera with scope mount or equivalent. The disturbance may be introduced as a load step output from a nominal setting within the designated tolerances of such a magnitude as not to exceed the maximum or minimum output load specified in 3.5.2. All instrumentation to be isolated from power line ground. At ambient temperatures that exceed the maximum allowable operating temperature or in the event inadequate cooling is provided, the thermostat (8849206) contacts will "make", thus removing input power to the module and simultaneously lighting the overheat light connected to terminals D and R, Figure 1. As an operational check the cooling air may be removed thus allowing the thermostat to attain the "make" temperature for which it is calibrated. Operation may be restored by allowing the module to cool and by depressing the reset switch. During the operational check the temperature of the edge of the cooler plate should be monitored at a position directly above the thermostat using a temperature potentiometer, Brown Instrument Type 126W2 or equivalent. At no time during the functional test should the plate edge be allowed to exceed 195°F before the overheat light operates.

4.4.2.1 Continuity check. There shall be continuity, not exceeding 0.5 ohm (resistive), between the pins indicated below. Continuity will be checked by a Keithley Milliohmmeter, model 502 or equivalent.

Terminals:	A, B, and C
Terminals:	R, P, N, K, L and M
Terminals:	Y, AA and CC
Terminals:	BB, DD and Z

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4.4.2.2 Insulation check. The insulation resistance shall be greater than 20 megohms between each of the following groups of terminals and case (measured each to the other two) using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma. for this test.

Group I - H,A,D,R, and Y tied together.
Group II - Unused Terminals - E,F,J,S,T,U,V,W and X tied together.

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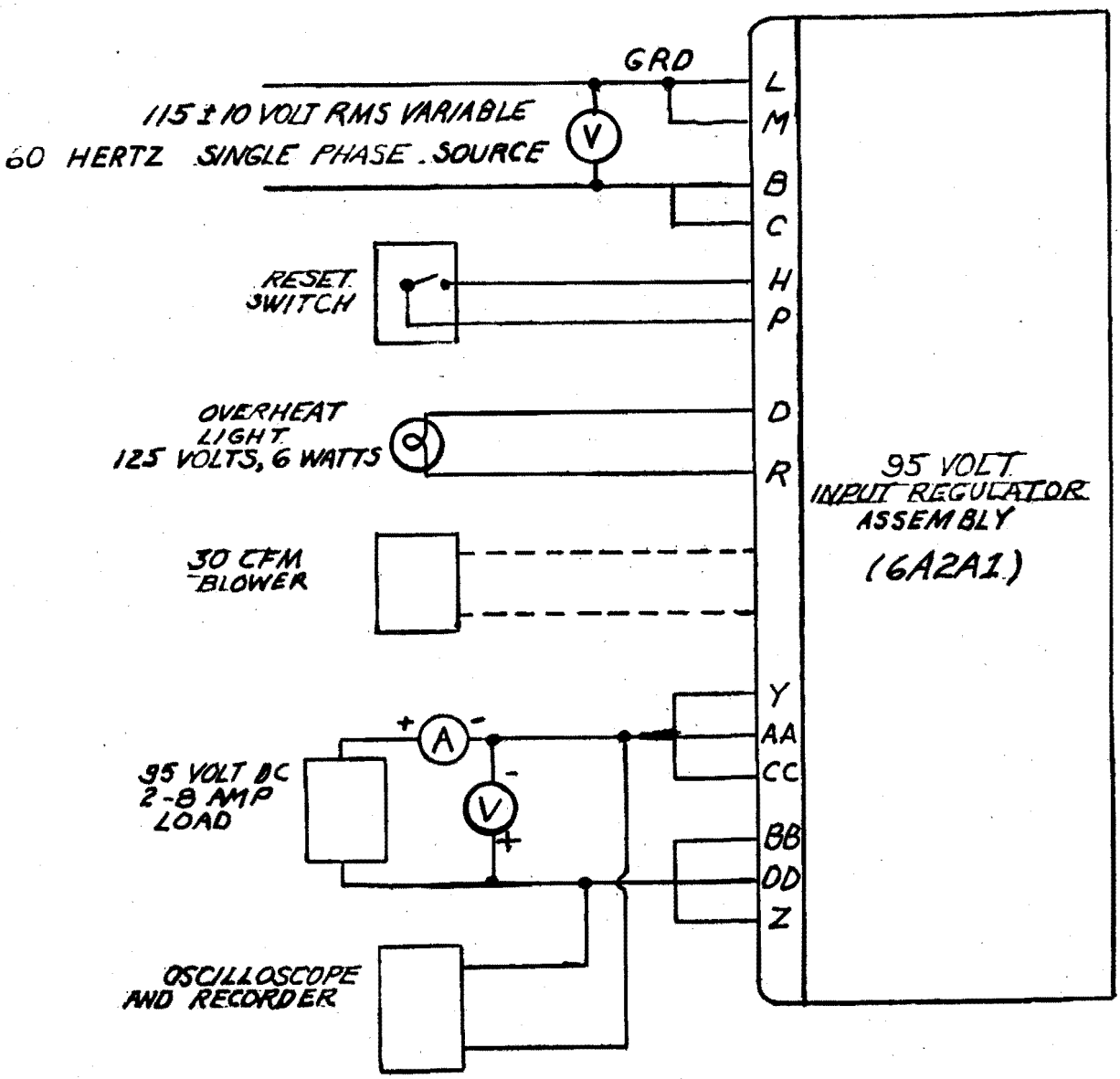


Figure 1 - Suggested test circuit for 95 volt input 6A2A1 regulator assembly

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APPENDIX C

1000 Cycle Inverter No. 1 Assembly

(6A2A2)

The following sections apply specifically to the 1000 Cycle Inverter No. 1 Assembly module. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-P- (MU).

3.5 Operational characteristics.
3.5.1 Calibration capability. Not applicable.
3.5.2 Operational limits. With the 6A2A2 Inverter Assembly connected as shown in Figure 1 of this appendix and with the input voltage varied as indicated, the following circuit limits will be maintained. This test shall be performed in accordance with 4.4.2.

Terminals

A-B	Input Voltage Variation	95 ± 1.9 volts dc
A-B	Input Voltage Ripple	1.0 volts rms (max.)
UVS-TRP	Terminal Output Voltage	16.5 ± 0.65 volts rms, 1100 ± 200 Hz
	Output Current Variation	10 ± 2 amps, ac
N-M	Terminal Output Voltage	20 ± volt, rms 1100 ± 200 Hz
	Output Current Variation	100 ± 10 ma, ac
KL & LJ	Terminal Output Voltage	20 + volts, rms 1100 ± 200 Hz
	Output Current Variation	60 ± 5 ma, ac
H-E	Terminal Output Voltage	450 ± 23 volts, rms, 1100 ± 200 Hz
	Output Current Variation	140 to 250 ma, ac

4.3.2.1 Classification of characteristics.
4.3.2.1.1 1000 Cycle Inverter No. 1 Assembly (6A2A2) (see dwg 8848035).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect location and orientation of connector	Visual
	102. Loose or missing fasteners	Visual/Manual
	103. Defective solder joints	Visual/Manual

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104. Component missing.....Visual
105. Cracked or broken connectors or damaged pins.....Visual
106. Board mounting screws loose.....Manual
107. Objective evidence relative to material improper or missing (see 3.1).....Visual
108. Serialization improper or missing (see 3.4).....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual
202. Component identification markings illegible or obliterated...Visual
203. Improper finish or color.....Visual

4.3.2.1.2 Preparation for delivery (see dwg 8883499).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Packing material and packaging not proper or missing	Visual
	102. Marking not in accordance with dwg	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.
4.4.1 Calibration capability. Not applicable.
4.4.2 Operational limits. Connect module as shown in Figure 1 of this appendix and make input and load adjustments as indicated in 3.5.2, this appendix. The dc input voltage will be measured with a Weston, model 931, dc voltmeter or equivalent. The ac output voltage will be measured with a Ballantine, model 320, ac vtvm or equivalent. Ac output currents will be measured with a Weston, Type V-C, Vacuum Thermoelement and a Millivac, model MV-17-C dc vtvm or equivalent. Frequency will be measured with a Hewlett-Packard Counter Model 521C or equivalent. All instrumentation will be isolated from power line ground.

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4.4.2.1 Continuity check. There shall be continuity, not exceeding 0.5 ohms (resistive) between the connector pins indicated below. Continuity will be checked with a Keithley milliohmmeter, model 502 or equivalent.

Terminals: U,V and S
Terminals: T,R and P

4.4.2.2 Insulation check. The insulation resistance shall be greater than 20 megohms between each of the following terminals and all others listed and case using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma. for this test.

Terminals: A, U, N, K and H
The unused terminals C, D and F shall be tied together and treated as one terminal.

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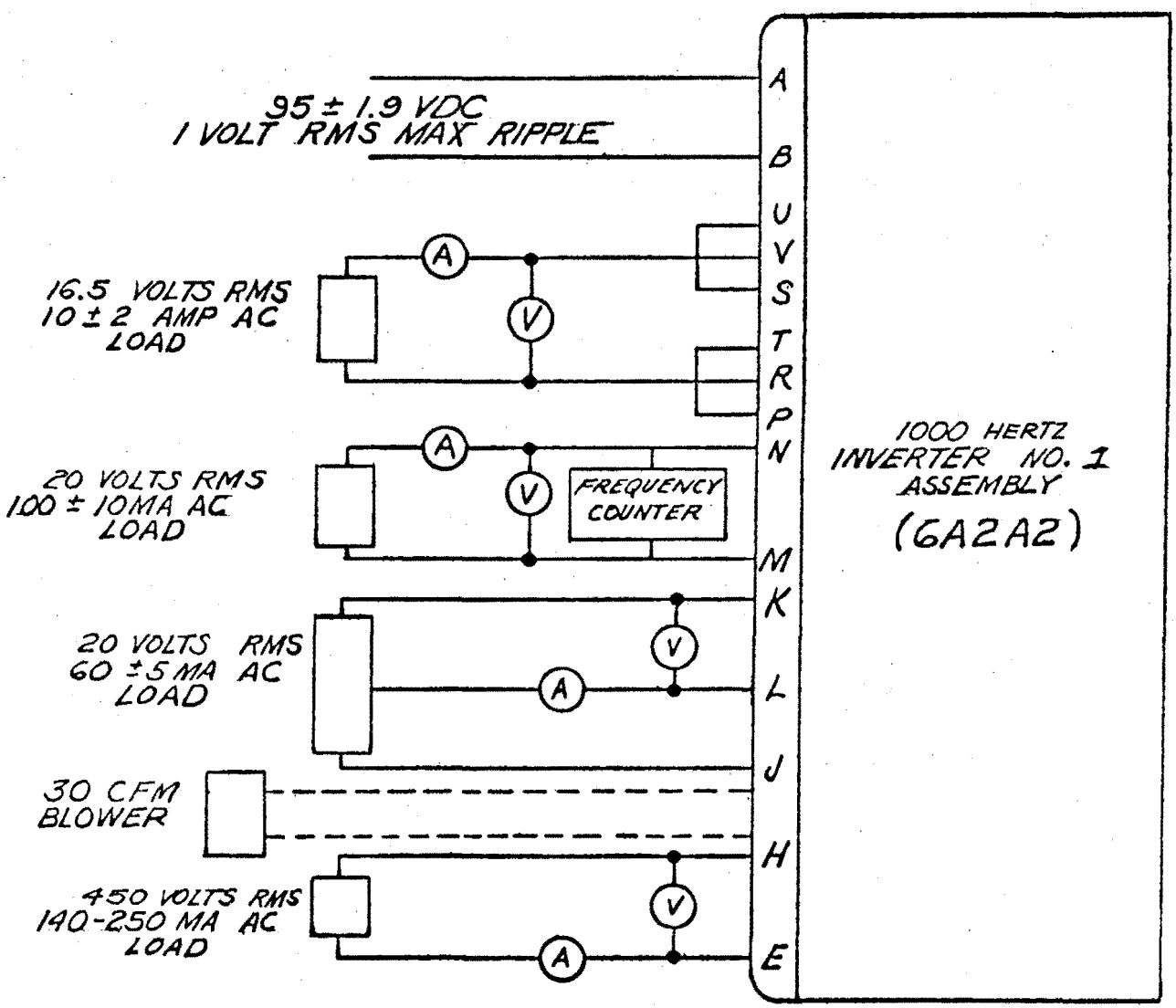


Figure 1 - Suggested test circuit for 1000 hertz inverter 6A2A3 No. 1 assembly

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APPENDIX D

1000 Cycle Inverter No. 2 Assembly

(6A2A3)

The following sections apply specifically to the 330-380 Volt Regulator Assembly module. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-P- (MU).

3.5 Operational characteristics.
3.5.1 Calibration capability. Not applicable.
3.5.2 Operational limits. With the 6A2A3 Inverter Assembly connected as shown in Figure 1 of this appendix and with the input voltage varied as indicated, the following circuit limits will be maintained:

Terminals

A-B	Input Voltage Variation	95 ± 1.9 volts dc
A-B	Input Voltage Ripple	1.0 volts rms (max.)
U-V	Terminal Output Voltage	520 ± 30 volts, rms, 1100 ± 200 Hz
	Output Current Variation	60 ± 10 ma, ac
ST & RP	Terminal Output Voltage	31.5 ± 1.5 volts, rms, 1100 ± 200 Hz
	Output Current Variation	3.5 to 8 amps, ac
N-M	Terminal Output Voltage	60 ± 3 volts, rms, 1100 ± 200 Hz
	Output Current Variation	80 ± 10 ma, ac
K-L	Terminal Output Voltage	68 ± 4 volts,rms, 1100 ± 200 Hz
	Output Current Variation	25 to 325 ma, ac
J-H	Terminal Output Voltage	40 ± 2.5 volts,rms 1100 ± 200 Hz
	Output Current Variation	350 ± 25 ma, ac

This test shall be performed in accordance with 4.4.2.

4.3.2.1 Classification of characteristics.
4.3.2.1.1 1000 Cycle Inverter No. 2 Assembly (6A2A3) (see dwg 8848055).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Incorrect location and orientation of connector.....Visual

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102. Loose or missing fasteners.....Visual/Manual
103. Defective solder joints.....Visual/Manual
104. Component missing.....Visual
105. Cracked or broken connectors or damaged pins.....Visual
106. Board mounting screws loose.....Manual
107. Objective evidence relative to material improper or missing (see 3.1).....Visual
108. Serialization improper or missing (see 3.4).....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual
202. Component identification markings illegible or obliterated.....Visual
203. Improper finish or color.....Visual

4.3.2.1.2 Preparation for delivery (see dwg 8883499).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Packing material and packaging not proper or missing	Visual
	102. Marking not in accordance with dwg	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.
4.4.1 Calibration capability. Not applicable.
4.4.2 Operational limits. Connect module as shown in Figure 1 of this appendix and make input and load adjustments as indicated in 3.5.2, this appendix. The dc input voltage will be measured with a Weston, model 931, dc voltmeter or equivalent. The ac output voltages will be measured with a Ballantine, model 320 vtvm or equivalent. The 1000 Hz output currents to be measured with a Type VI, Weston vacuum thermo-element with a Millivac, model MV-17-C,dc, vtvm or equivalent. Frequency will be measured with a Hewlett-Packard Counter, model 521C or equivalent. All instrumentation will be isolated from power line ground.

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4.4.2.1 Continuity check. There shall be continuity, not exceeding 0.5 ohms (resistive) between the connector pins indicated below. Continuity will be checked with a Keithley milliohmmeter, model 502.

Terminals: S and T
Terminals: R and P

4.4.2.2 Insulation check. The insulation resistance shall be greater than 20 megohms between each of the following terminals and all others listed and case using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma. for this test.

Terminals: A, U, S, N, K, and J.
The unused Terminals C, D, E, R, and H shall be tied together and tested as one terminal.

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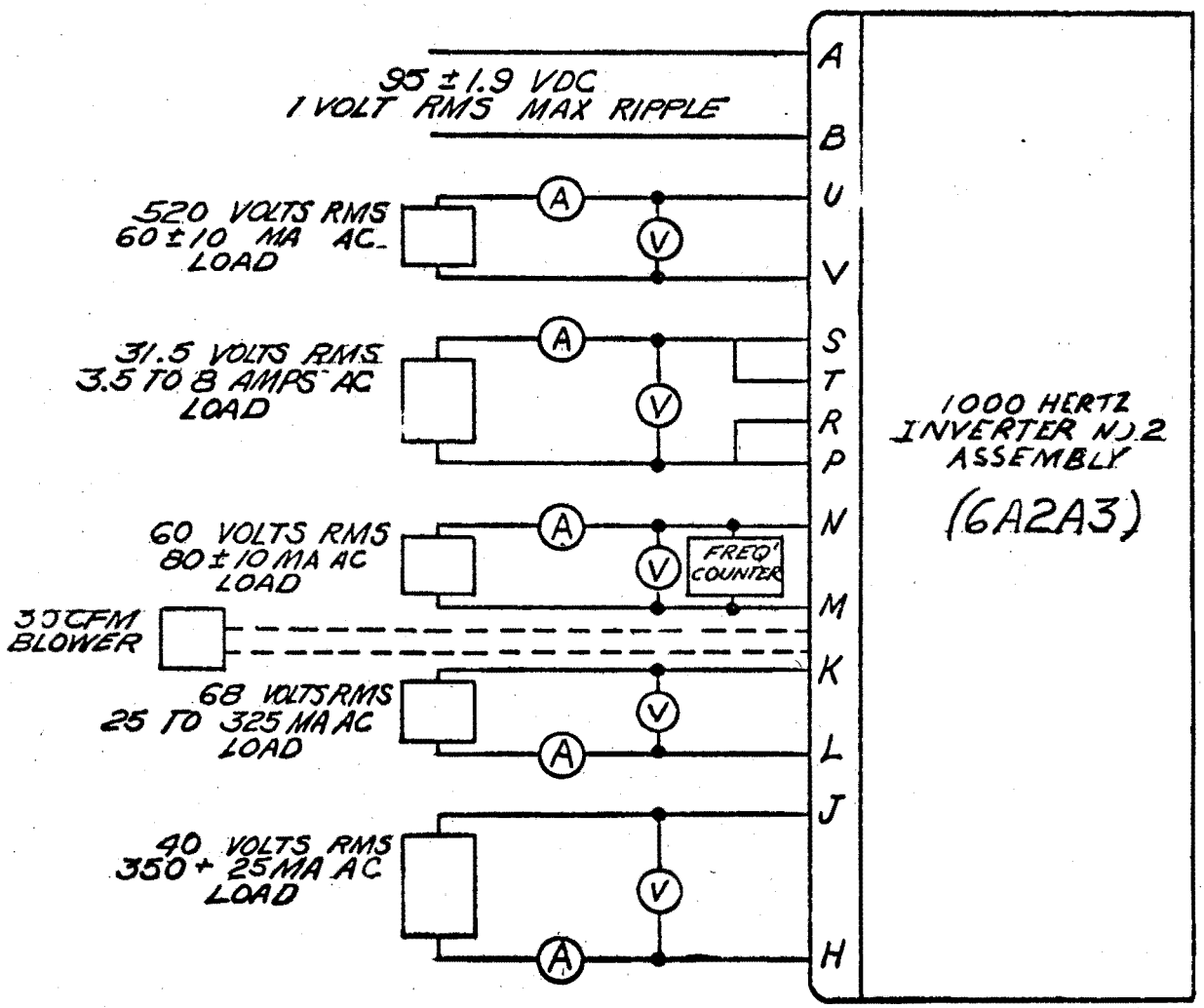


Figure 1 - Suggested test circuit for the 1000 hertz inverter 6A2A3 No. 2 assembly

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APPENDIX E

330-380 Volt Regulator Assembly

(6A2A4)

The following sections apply specifically to the 330-380 Volt Regulator Assembly module. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-P- (MU).

3.5 Operational characteristics.

3.5.1 Calibration capability. Under all conditions indicated in 3.5.2 below, operation of calibration pots R7 and R10 shall provide output voltage variation, at rated maximum load, as shown below. The calibration shall be performed in accordance with 4.4.1.

Relay K1: Contacts 4-3 & 8-6 Made

Output Voltage	375 volts to 385 vdc

Relay K1: Contacts 4-5 & 8-7 Made

Output Voltage	325 volts to 335 vdc

Nomimal settings for final calibration to be 330 volts at 175 ma load current and 380 volts at 225 ma load current for the relay positions stipulated above.

3.5.2 Operational limits. With the 6A2A4 Regulator Assembly connected as shown in Figure 1 of this appendix and with the Input voltage and load current varied as indicated, the following circuit limits will be maintained. This test shall be performed in accordance with 4.4.2.

Terminals

A-C	Input Voltage Variation	450 ± 23 volts, rms, 1100 ± 200 Hz

Relay K1: Contacts 4-3 & 8-6 Made (coil not energized)

R & P	Output Voltage	380 ± 7.6 volts dc
	Output Current Variation	205 to 240 ma, dc
	Regulation	± 2%
	Ripple Voltage	500 mv peak (max)

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	Response	Output Voltage to be within regulation limits within 1.0 millisecond after disturbance.

Relay K1: Contacts 4-5 & 8-7 Made (coil energized)

Output voltage	330 ± 6.6 volts dc
Output current variation	150 to 195 ma, dc
Regulation	± 2%
Ripple voltage	500 mv peak (max)
Response	Output voltage to be within regulation limits within 1.0 millisecond after disturbance.

4.3.2.1 Classification of characteristics.

4.3.2.1.1 330-380 Volt Regulator Assembly (6A2A4) (see dwg 8848094).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector	Visual
102.	Loose or missing fasteners	Visual/Manual
103.	Defective solder joints	Visual/Manual
104.	Component missing	Visual
105.	Cracked or broken connectors or damaged pins	Visual
106.	Board mounting screws loose	Manual
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual
108.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual
202.	Component identification markings illegible or obliterated	Visual
203.	Improper finish or color	Visual

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4.3.2.1.2 Preparation for delivery (see dwg 8883500).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Packing material and packaging not proper or missing	Visual
	102. Marking not in accordance with dwg	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.

4.4.1 Calibration capability. Connect module as shown in Figure 1 of this appendix and make the adjustments indicated in 3.5.1, this appendix. The output may be manually adjusted for the nominal output voltages by turning the lower calibration pot, R10, clockwise to increase the 380 volt output, and after applying 28 volts dc to terminals K and H, by adjusting the upper calibration pot, R7, clockwise to increase the 330 volt output. Dc voltages to be measured with a Weston, model 931 dc voltmeter or equivalent. Dc to be measured with a Weston model 45 dc ammeter with a 0-0.5 amp Weston, model 45 shunt or equivalent. Ac input voltage to be measured with a Ballantine, model 320 vtvm or equivalent. All instrumentation to be isolated from power line ground. The assembly shall be calibrated at a temperature of 80 ± 30°F.

4.4.2 Operational limits. Connect module as shown in Figure 1 of this appendix. Throughout the input and output variations specified, the module must stay within the limits described in 3.5.2, this appendix. Ac input voltages and dc output voltages and currents are to be measured as prescribed in 4.4.1 above. Ripple voltage and response time to be measured and recorded with a Hewlett-Packard, model 130B oscilloscope and Polaroid camera. Transient voltage changes of less than 25 microseconds duration shall not be considered during ripple voltage measurements. The disturbance may be introduced as a load step output from a nominal setting, and will be of such magnitude as not to exceed the minimum output load limits specified in 3.5.2. All instrumentation to be isolated from power line ground.

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4.4.2.1 Insulation check. The insulation resistance shall be greater then 20 megohms between each of the following terminals and all others listed and case using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma for this test.

Terminals: E, H, J and P.
The unused terminals B, D, F, L, M and N shall be tied together and treated as one terminal.

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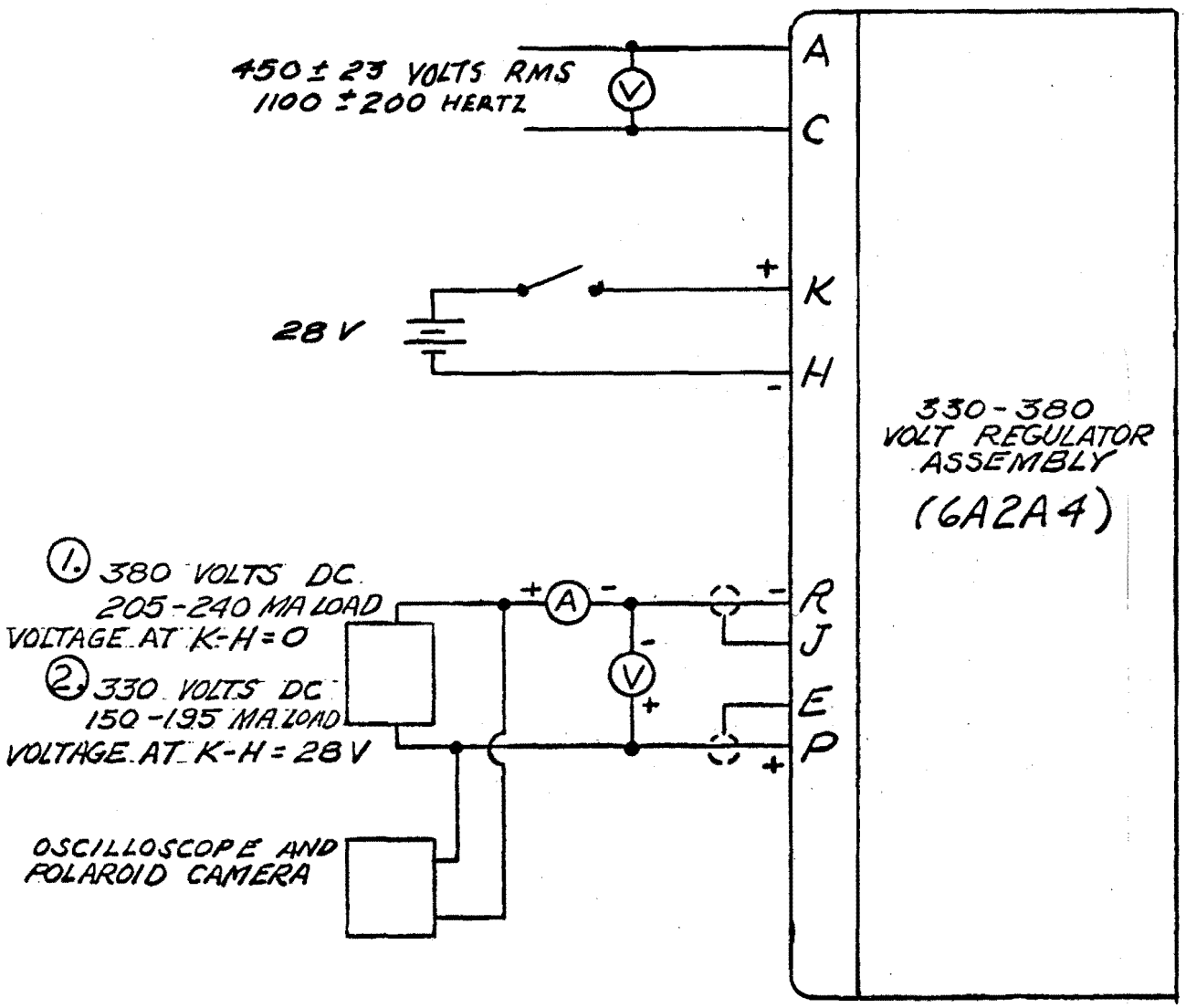


Figure 1 - Suggested test circuit for the 330-380 volt regulator (6A2A4) assembly

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APPENDIX F

6.0 & 7.0 Volt Regulator Assembly

(6A2A5)

The following sections apply specifically to the 6.0 and 7.0 Volt Regulator Assembly module. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-P- (MU).

3.5 Operational characteristics.

3.5.1 Calibration capability. Under all conditions indicated in 3.5.2 below and with relay K1 operated to make contacts 7 and 6, operation of calibration pot R14 shall provide output voltage adjustment, at 9.2 amps load current, greater than 6.1 volts and less than 5.9 volts. With contacts 7 and 5 of relay K1 made, operation of calibration pot R12 shall provide output voltage adjustment at 10.25 amps load current, greater than 7.1 and less than 6.9 volts. Final calibration to be performed in the above sequence. The calibration shall be performed in accordance with 4.4.1.

3.5.2 Operational limits. Connect the 6A2A5 Regulator Assembly as shown in Figure 1 of this appendix. With the input voltages and output current varied as indicated below, the following circuit limits will be maintained. This test shall be performed in accordance with 4.4.4.

Terminals

ABC-DEF	Input voltage variation	16.5 ± 0.65 volts,rms square wave, 1100 ± 200 Hz
H-J	Input voltage variation	20 ± 1 volt, rms, square wave, 1100 ± 200 Hz
K-L-M	Input voltage variation	40 ± 2 volts, rms, square wave 1100 ± 200 Hz

7.0 volt section

Relay K1, Contacts 7-5 Made; Coil (pin R,U) energized

Terminals

MM NN LL-JJ FF HH	Output current variation	9.5 to 11 amps,
MM NN LL-JJ FF HH	Output voltage	7.0 ± 0.14 volts dc
	Ripple voltage	100 mv peak (max)

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	Regulation	± 2%
	Response	Output voltage to be within regulation limits within 1.0 milliseconds after disturbance.

6.0 Volt Section

Relay K1, contacts 7-6 made; coil (pin R, N) energized

Terminals

MM NN LL-JJ FF HH	Output current variation	8.7 to 9.7 amps,
MM NN LL-JJ FF HH	Output voltage	6.0 ± 0.12 volts dc
	Ripple voltage	100 mv peak (max)
	Regulation	± 2%
	Response	Output voltage to be within regulation limits within 1.0 milliseconds after disturbance.

4.3.2.1 Classification of characteristics.

4.3.2.1.1 6.0 and 7.0 volt regulator assembly (6A2A5) (see dwg 8833322).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector	Visual
102.	Loose or missing fasteners	Visual/Manual
103.	Defective solder joints	Visual/Manual
104.	Component missing	Visual
105.	Cracked or broken connectors or damaged pins	Visual
106.	Board mounting screws loose	Manual
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual
108.	Serialization improper or missing (see 3.4)	Visual

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Minor:
201. Evidence of poor workmanship (see 3.8).....Visual
202. Component identification markings illegible or obliterated.....Visual
203. Improper finish or color.....Visual

4.3.2.1.2 Preparation for delivery (see dwg 8883501).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.

4.4.1 Calibration capability. Connect module as shown in Figure 1 of this appendix and make the adjustments specified in 3.5.1, this appendix. The module may be manually adjusted for a nomimal output voltage by turning the pot shafts clockwise to increase the voltages. Rotation of R12 shaft will provide adjustment for the 7.0 volts section when 28 volts dc is applied to pins R and U. Rotation of R14 will provide adjustment for the 6.0 volts section when 28 volts dc is applied to pins R and N. Ac voltage readings shall be made with a Hewlett-Packard, Model 400 H ac vtvm or equivalent. Dc voltage readings shall be made with a Weston, Model 931, dc voltmeter or equivalent and dc current shall be measured with a Weston, Model 45, dc ammeter with a 0-20 amp Weston, Model 45 shunt or equivalent. All instrumentation to be isolated from power line ground. The assembly shall be calibrated at a temperature of 80 ± 30°F.

4.4.2 Operational limits. Connect the module as indicated in Figure 1 of this appendix and make the adjustment specified in 3.5.2, this appendix. Voltage and current readings will be measured with the instrumentation described in 4.4.1, this appendix. Ripple voltage and response time to be measured with a Hewlett-Packard, Model 130B oscilloscope. Transient voltage changes of less than 25 microseconds duration shall not be considered during ripple voltage measurements. The disturbance may be introduced as a load step

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output from a nominal setting within the designated tolerances and will be of such magnitude as not to exceed the minimum or maximum output load limits specified in 3.5.2. All instrumentation to be isolated from power line ground.

4.4.2.1 Continuity check. There shall be continuity not exceeding 0.5 ohms (resistive) between the pins indicated below. Continuity will be checked by a Keithley Milliohmmeter, Model 502 or equivalent. Apply 28 ± 2 vdc between pins N and R before testing.

Terminals: A, B and C
Terminals: D, E and F
Terminals: MM, NN and LL
Terminals: JJ, FF and HH
Terminals: T and P.

4.4.2.2 Insulation check. The insulation resistance shall be greater than 20 megohms between each of the following terminals and all others listed and case using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma for this test. Momentarily apply 28 ± 2 volts dc between terminals U (positive) and R (negative) before testing.

Terminals: *A, EE, W, T, P, and U.
The unused terminals S, V, X, Y, Z, BB, CC, and DD shall be tied together and tested as one terminal.

* Positive megohmmeter lead.

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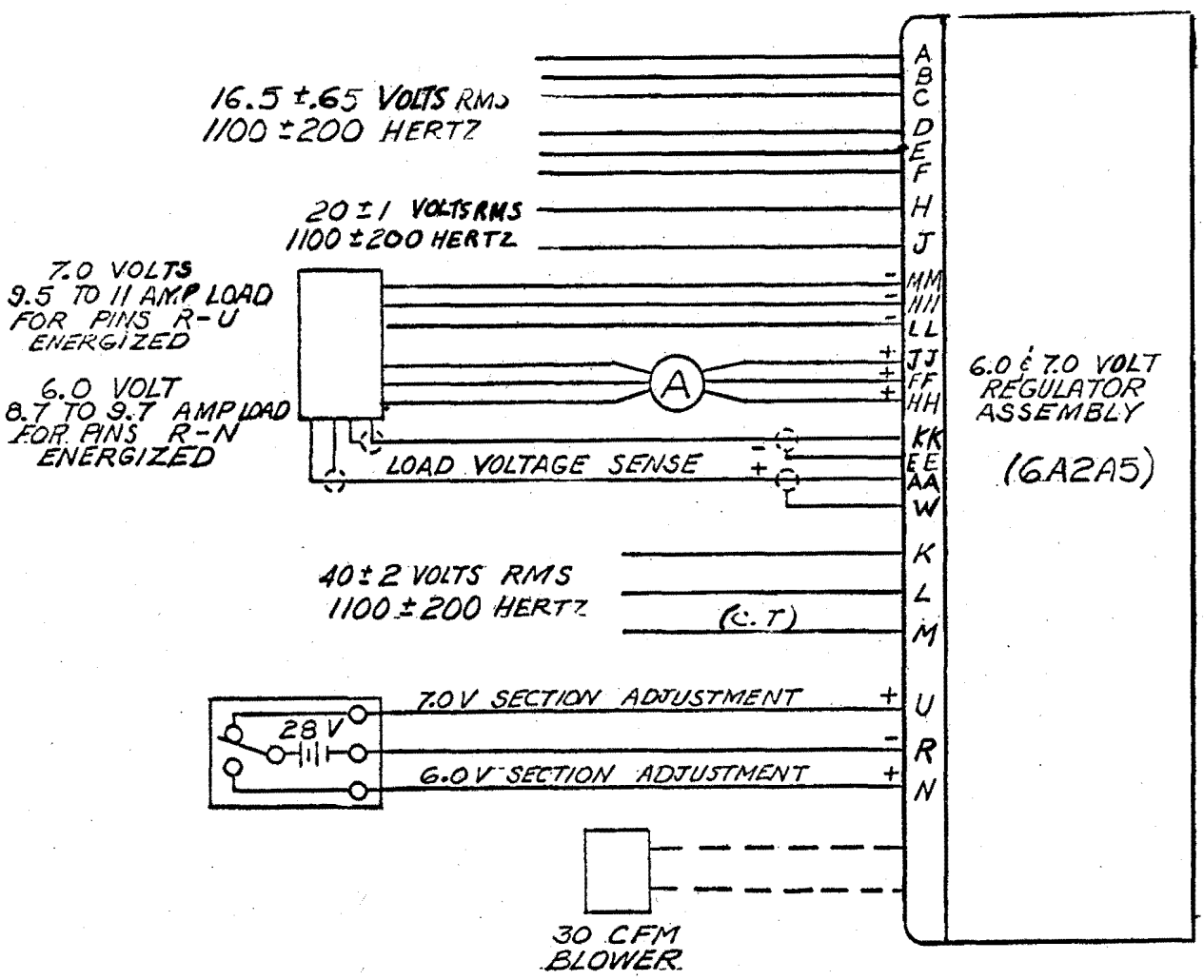


Figure 1 - Suggested test circuit for the 6.0 and 7.0 volt 6A2A5 regulator assembly

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APPENDIX G

350, -10, -28, 56 Volt Regulator Assembly

(6A2A6)

The following sections apply specifically to the 350, -10, -28, 56 Volt Regulator Assembly module. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-P- (MU).

3.5 Operational characteristics.

3.5.1 Calibration capability. Under all conditions indicated under the appropriate section in 3.5.2 below, operation of calibration pot R13 of the 56 volt section shall provide output voltage adjustment, at a 150 ma load current, greater than 56.5 volts dc and less than 55.5 volts dc. The 350, 28, and 10 volt sections are fixed and offer no calibration adjustment. The calibration shall be performed in accordance with 4.4.1.

3.5.2 Operational limits. With the 6A2A6 Regulator Assembly connected as shown in Figure 1 of this appendix and with the input voltage and load current varied as indicated below, the following circuit limits will be maintained:

-350 Volt Section

Terminals		
A-B	Input voltage variation	520 ± 30 volts, rms, square wave 1100 ± 200 Hz
CC-DD	Output current variation	30 ± 10 ma,
CC-DD	Output voltage	350 ± 17.5 volts dc
	Regulation	± 5%
	Ripple voltage	360 mv rms (max)
	Response	Output voltage to be within regulation limits within 1.0 milliseconds after load disturbance.

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-56 Volt Section

Terminals

J-H	Input voltage variation	68 ± 4 volts, rms, square wave 1100 ± 200 Hz
SU-R	Output current variation	0-300 ma,
SU-R	Output voltage	56 ± 2 volts, dc
	Regulation	± 3.5%
	Ripple voltage	500 mv rms (max)
	Response	Output voltage to be within regulation limits within 1.0 milliseconds after disturbance.

The test shall be performed in accordance with 4.4.4.

-28 Volt Section

Terminals

E-F	Input voltage variation	60 ± 3 volts, rms, square wave 1100 ± 200 Hz
Z-V	Output current variation	15 ± 1.5 ma,
Z-V	Output voltage	28 ± 2.8 volts, dc
	Regulation	± 10%
	Ripple voltage	5 mv rms (max)
	Response	Output voltage to be within regulation limits within 1.0 milliseconds after disturbance.

-10 Volt Section

E-F	Input voltage variation	60 ± 3 volts, rms, square wave 1100 ± 200 Hz
Y-Z	Output current variation	30 ± 3 ma,
Y-Z	Output voltage	10 ± 1 volt, dc
	Regulation	± 10%
	Ripple voltage	5 mv rms (max)
	Response	Output voltage to be within regulation limits within 1.0 milliseconds after disturbance.

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4.3.2.1 Classification of characteristics.

4.3.2.1.1 350, -10, -28, 56 volt regulator assembly (6A2A6) (see dwg 8833323).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector	Visual
102.	Loose or missing fasteners	Visual/Manual
103.	Defective solder joints	Visual/Manual
104.	Component missing	Visual
105.	Cracked or broken connectors or damaged pins	Visual
106.	Board mounting screws loose	Manual
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual
108.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual
202.	Conponent identification markings illegible or obliterated	Visual
203.	Improper finish or color	Visual

4.3.2.1.2 Preparation for delivery (see dwg 8883502).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.

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4.4.1 Calibration capability. Connect module as shown in Figure 1 of this appendix and make the adjustments indicated in 3.5.1, this appendix. The module shall be manually adjusted for a nominal output voltage by turning the R13 pot shaft clockwise to increase the voltage. Ac voltage measurements will be made with a Hewlett-Packard Model 400H, ac, vtvm or equivalent. Dc voltage will be measured with a Weston Model 931, dc voltmeter or equivalent, dc current will be measured with a Weston Model 45 dc ammeter with a 0-0.5 amp, Weston Model 45 shunt or equivalent. All instrumentation to be isolated from power line ground. The assembly shall be calibrated at a temperature of 80 ± 30°F.

4.4.2 Operational limits. Connect module as indicated in Figure 1 of this appendix and make the adjustments specified in 3.5.2, this appendix. The ac input voltages are to be measured with a Hewlett-Packard, Model 400H, rms, vtvm or equivalent. Ripple to be measured with a Fluke, rms voltmeter, Model 910A, using the lowest scale setting possible without exceeding maximum scale limits. Response to be measured with a Hewlett-Packard, Model 130 B oscilloscope or equivalent and recorded by a Polaroid Land Camera with scope mount. The disturbance may be introduced as a load step output, magnitude as not to exceed the minimum or maximum output load limits specified in 3.5.2. The dc output voltages shall be measured with a Weston, Model 931, dc voltmeter or equivalent. The dc output currents shall be measured with a Weston, Model 931, dc milliammeter or equivalent. All instrumentation to isolated from power line ground.

4.4.2.1 Continuity check. There shall be continuity not exceeding 0.5 ohms between the terminals U and S. Continuity will be checked by a Model 502 Keithley milliohmmeter or equivalent.

4.4.2.2 Insulation check. The insulation resistance shall be greater than 20 megohms between each of the following terminals and all others listed and case using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma for this test.

Terminals: A, E, W, X, T, and J.

The unused terminals C, D, K, L, M, N, P, AA and BB shall be tied together and tested as one terminal.

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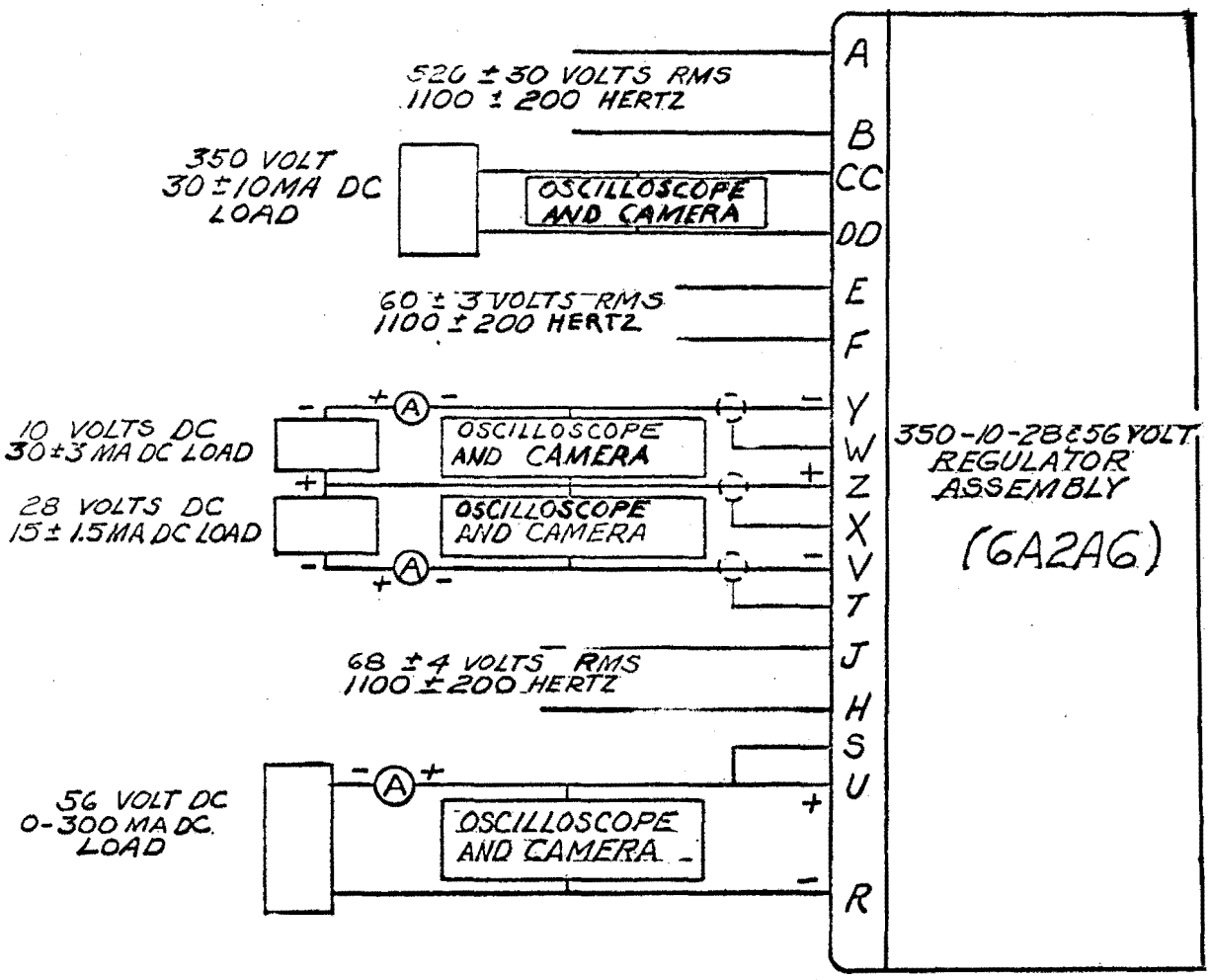


Figure 1 - Suggested test circuit for the 350, 10, 28 and 56 volt regulator assembly

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APPENDIX H

28 Volt Output Assembly

(6A2A7)

The following sections apply specifically to the 28 Volt Output Assembly module. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-P- (MU).

3.5 Operational characteristics.

3.5.1 Calibration capability. Under all conditions indicated under the appropriate section in 3.5.2 below, operation of calibration pot R10 in the series regulated section shall provide output voltage adjustment at a 350 ma load current, greater than 28.5 volts dc and less than 27.5 volts dc. The non-regulated section is fixed and offers no calibration adjustment. The calibration shall be performed in accordance with 4.4.1.

3.5.2 Operational limits. With the 6A2A7 output assembly connected as shown in Figure 1 of this appendix and with the input voltage and load current varied as indicated, the following circuit limits will be maintained. The test shall be performed in accordance with 4.4.2.

Series Regulated 28 Volt Supply

Terminals		
E-F	Input voltage variation	40 ± 2.5 volts, rms, square wave 1100 ± 200 Hz
CC-DD	Output current variation	350 ± 25 ma, dc
CC-DD	Output voltage variation	28 ± 0.84 volts, dc
	Regulation	± 3%
	Ripple voltage	10 mv, rms (max)
	Response	Output voltage to be within regulation limits within 1.0 millisecond after disturbance.

Non-Regulated 28 Volt Supply

Terminals		
AC-BD	Input voltage variation	31 ± 1.5 volts, rms, square wave, 1100 ± 200 Hz

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ZYXW-VUTSRP	Input voltage variation	3.5 ± 8.0 amps, dc
ZYXW-VUTSRP	Output voltage	29 ± 3 volts, dc
	Ripple voltage	1.0 volts, rms (max)

4.3.2.1 Classification of characteristics.

4.3.2.1.1 28 volt output assembly (6A2A7) (see dwg 8833324).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector	Visual
102.	Loose or missing fasteners	Visual/Manual
103.	Defective solder joints	Visual/Manual
104.	Component missing	Visual
105.	Cracked or broken connectors or damaged pins	Visual
106.	Board mounting screws loose	Manual
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual
108.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual
202.	Component identification markings illegible or obliterated	Visual
203.	Improper finish or color	Visual

4.3.2.1.2 Preparation for delivery (see dwg 8883503).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.

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4.4.1 Calibration capability. Connect module as shown in Figure 1 of this appendix and make the adjustments indicated in 3.5.1, this appendix. The regulated 28 volt section of this module shall be adjusted by turning the R10 pot shaft clockwise to increase output voltage. Ac voltage measurements will be made with a Hewlett-Packard, Model 400H, ac, vtvm or equivalent. Dc voltage will be measured with a Weston, Model 931, dc voltmeter or equivalent. Dc will be measured with a General Electric, Model DP-9, dc milliammeter or equivalent. All instrumentation to be isolated from power line ground. The assembly shall be calibrated at a temperature of 80 ± 30°F.

4.4.2 Operational limits. Connect module as indicated in Figure 1 of this appendix and make the adjustments specified in 3.5.2, this appendix. The ac input voltages are to be measured with a Hewlett-Packard, Model 400H, ac vtvm or equivalent. Ripple to be measured with a Fluke, rms voltmeter, Model 910 A or equivalent. Response of output voltage is to be measured with a Hewlett-Packard, Model 130B, oscilloscope or equivalent and recorded by a Polaroid Land Camera with scope mount. The disturbance may be introduced as a load step output, from a nominal setting within the designated tolerances and will be of such magnitude as not to exceed the maximum or minimum output load limits specified in 3.5.2. All instrumentation to be isolated from power line ground. The outputs of the two sections ar to be measured as follows:

Regulated 28 Volt Section

Terminals

CC-DD	Output voltage:	Weston, Model 931 dc voltmeter or equivalent
CC-DD	Output current:	General Electric, Model DP-9 dc milliammeter or equivalent.

Non-Regulated 28 Volt Section

ZYXW-VUTSRP	Output voltage:	Weston, Model 931 dc volt= meter or equivalent
ZYXW-VUTSRP	Ouptut current:	Weston, Model 45 dc ammeter with a 0-10 amp Model 45 shunt or equivalent.

4.4.2.1 Continuity check. There shall be continuity not exceeding 0.5 ohms (resistive) between the connector pins indicated below. Continuity will be checked with Keithley milliohmmeter, Model 502.

Terminals: A and C
Terminals: B and D
Terminals: Z, Y, X and W
Terminals: V, U, T, S, R and P

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4.4.2.2 Insulation check. The insulation resistance shall be greater than 20 megohms between each of the following terminals and all others listed and case using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma for this test.

Terminals: *V, DD, AA and BB.
*Positive megohmmeter lead.
The unused terminals H, J, K, L, M and M shall be tied together and tested as one terminal.

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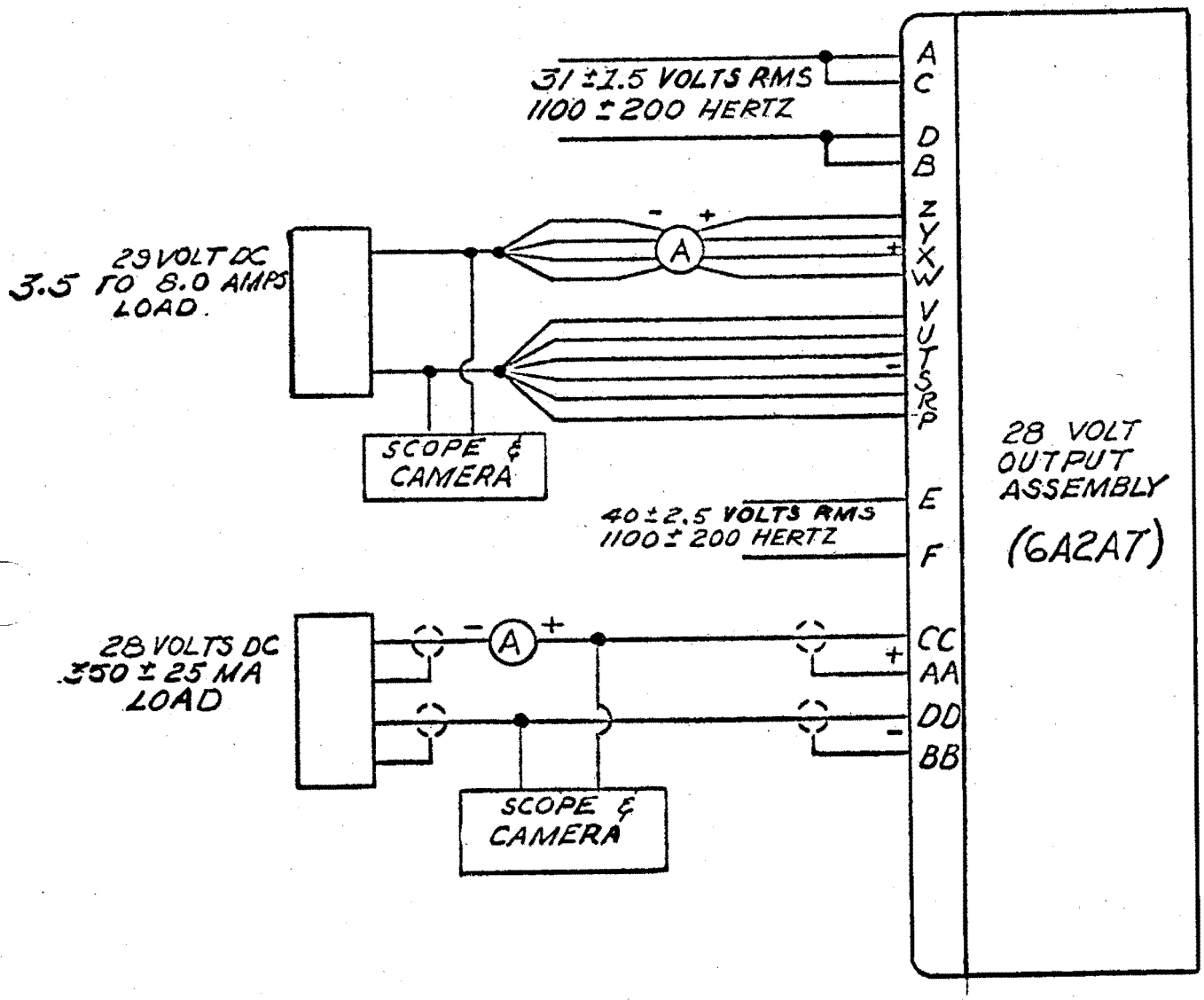


Figure 1 - Suggested test circuit for the 28 volt 6A2A7 output assembly

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APPENDIX I

400 Cycle Inverter Assembly

(6A2A8)

The following sections apply specifically to the 400 Cycle Inverter Assembly module. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-P- (MU).

3.5 Operational characteristics.

3.5.1 Calibration capability. Under all conditions indicated under 3.5.2 below, operation of calibration pot R6 shall provide output voltage adjustment at a 250 ma load current, greater than 117 volts and less than 113 volts. For final calibration R6 is to be adjusted so as to provide 115 volts, 400 ± 7 Hz at 250 ma at terminals R and P. This calibration shall be performed in accordance with 4.4.1.

3.5.2 Operational limits. With the 6A2A8 Inverter Assembly connected as shown in Figure 1 of this appendix and with the input voltage and load current varied as indicated, the following circuit limits will be maintained. This test shall be performed in accordance with 4.4.2.

Terminals

A-C	Input voltage variation	29 ± 3 volts, dc 1 volt rms max ripple
R-P	Output current variation	250 ± 50 ma, ac
N-L	Output current variation	130 ± 10 ma, ac
R-P	Output voltage	115 ± 5.75 volt, rms, 400 ± 20 Hz
N-L	Output voltage	6.3 ± 0.315 volts, rms, 400 ± 20 Hz
	Regulation	± 5%
	% Harmonic	Not critical

3.5.2.1 Output phase relation. The output voltage from terminal R to terminal P shall be in phase with the output voltage from terminal N to terminal L. Testing shall be performed in accordance with 4.4.2.2.

4.3.2.1 Classification of characteristics.

4.3.2.1.1 400 Cycle Inverter Assembly (6A2A8) (see dwg 8833325).

Categories	Characteristics	Method of Inspection
Critical: None defined.		

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Major:

101.	Incorrect location and orientation of connector	Visual
102.	Loose or missing fasteners	Visual/Manual
103.	Defective solder joints	Visual/Manual
104.	Component missing	Visual
105.	Cracked or broken connectors or damaged pins	Visual
106.	Board mounting screws loose	Manual
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual
108.	Serialization improper or missing (see 3.4)	Visual

Minor:

201.	Evidence of poor workmanship (see 3.8)	Visual
202.	Component identification markings illegible or obliterated	Visual
203.	Improper finish or color	Visual

4.3.2.1.2 Preparation for delivery (see dwg 8883504).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.

4.4.1 Calibration capability. Connect the module as shown in Figure 1 of this appendix and make the adjustments specified in 3.5.1, this appendix. Output voltages shall be adjusted by turning the R6 pot shaft clockwise to increase output voltage. Ac voltages are to be measured by Weston, Model 433, ac milliammeters or equivalent. Input dc voltage to be measured by a General Electric, Model DP-9, dc voltmeter or equivalent. All instrumentation to be isolated from power line ground. The assembly shall be calibrated at a temperature of 80 ± 30°F.

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4.4.2 Operational limits. Connect the module as shown in Figure 1 of this appendix and make the adjustments specified in 3.5.2, this appendix. Voltage and current readings will be measured with the instrumentation described in 4.4.1, this appendix. Transient response and distortion to be measured with a Hewlett-Packard, Model 130B oscilloscope and recorded by a Polaroid Land Camera with scope mount. Frequency to be measured by a Hewlett-Packard, Model 521C, 5 place frequency counter or equivalent. All instrumentation to be isolated from power line ground.

4.4.2.1 Insulation check. The insulation resistance shall be greater than 20 megohms between each of the following terminals and all others listed and case using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma for this test.

Terminals: *A, R and N
*Positive megohmmeter lead.
The unused terminals B, D, E, H, J, K and M shall be tied together and tested as one terminal.

4.4.2.2 Output phase relation. Connect the assembly as shown in Figure 1 of this appendix and adjust the input voltage and output current to the values specified below. Connect terminals P and N together and the output voltage shall be as specified.

Terminals

A-C	Input voltage	29 ± 1 volts,dc
R-P	Output current	250 ± 10 ma, ac
N-L	Output current	130 ± 1- ma, ac
R-L	Output voltage	121.3 ± 6.2 volts, rms

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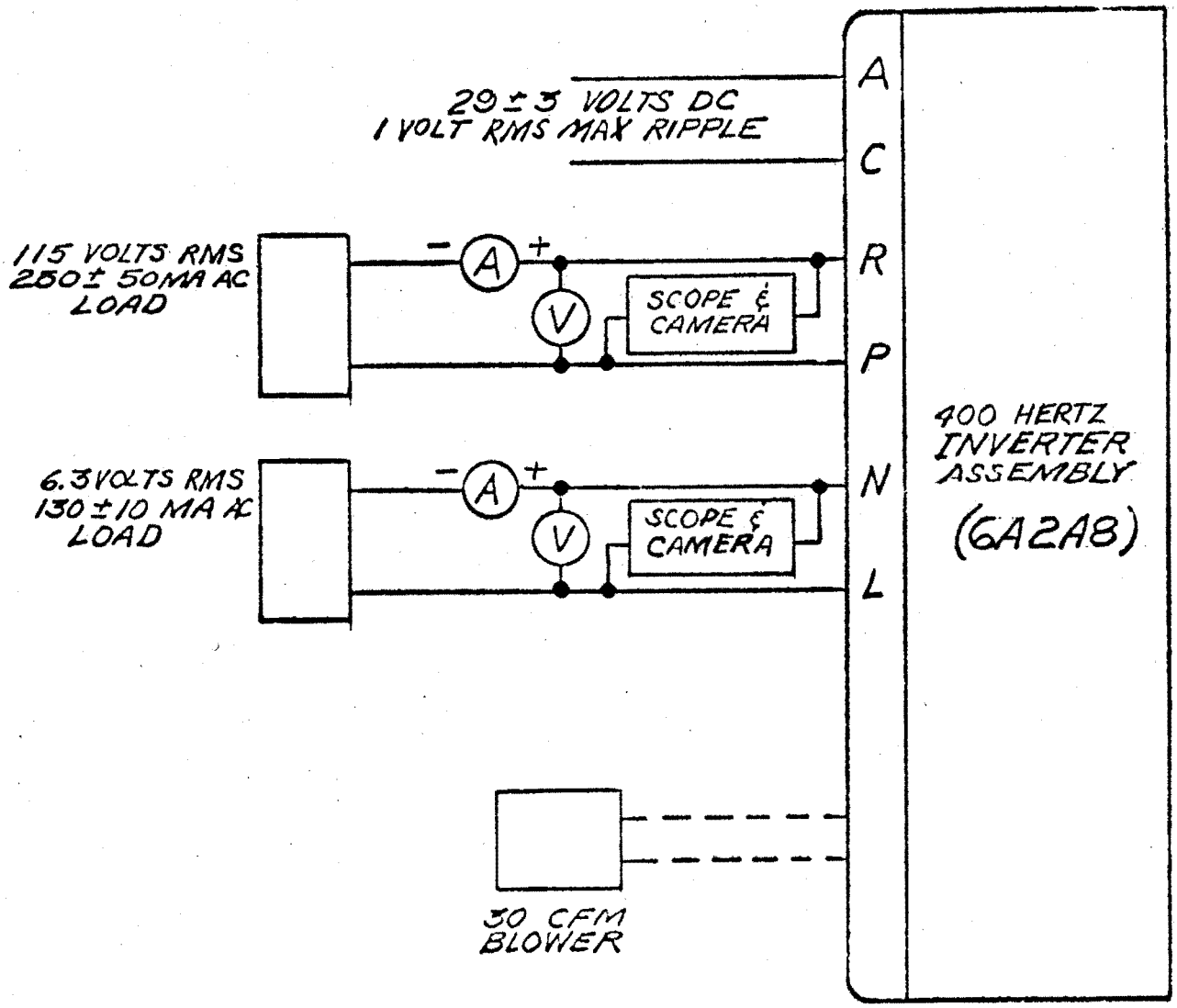


Figure 1 - Suggested test circuit for the 400 hertz 6A2A8 inverter assembly

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APPENDIX J

Rack and Panel Assembly

(6A2A9)

The following sections apply specifically to the power supply auxillary equipment portion of the Rack and Panel Assembly which is composed of the Fan (8886903), the Solenoid Valve, type V51 (8849230), the Pressure Switch (8847621), the Rotary Vacuum Pump (8847624) and the Reset Switch (8849250). In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-P- (MU).

3.5 Operational characteristics

3.5.1 Calibration capability. Not applicable.

3.5.2 Operational limits. The following functional checks will be performed:

3.5.2.1 Fan (8886903). The fan shall operate with a 115 ± 10 volt, 60 Hz source applied to pins r and W of connector P1, with N and M jumpered on J1.

3.5.2.2 Solenoid value (8849230), pressure switch (8847621) and rotary vacuum pump (8847624). There shall be continuity, within an interval of 10 seconds, between socket J6-S and pin P1-EE, with 115 ± 10 volts, 60 Hz, connected to socket J1-K and P1-W, and with 29 ± 3 vdc connected between socket J7-T (negative), and pins P1-FF and P1-DD (positive).

3.5.2.3 Probe adaptor assembly (8857106). With the probe adaptor assembly connected to the 6A2A9 vacuum inlet, there shall be continuity, within an interval of 10 seconds, between socket J6-S and pin P1-EE, with 115 ± 10 volts, 60 Hz, connected to socket J1-K and pin P1-W, with 29 ± 3 vdc connected positive to P1-FF, negative to J7-T, and with the opening of the probe sealed with its protective plug (8858063). With the probe unsealed there shall be no continuity between socket J6-S and pin P1-EE.

3.5.2.4 Reset switch (8849250). With the reset switch button depressed, there shall be continuity between sockets J1-P and J1-H.

3.5.3 Continuity and resistance. Continuity and resistance shall be within the specified limits between "primary" and "secondary" terminals listed in Table I. Testing shall be performed as specified in 4.4.2.1.

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3.5.4 Insulation resistance. The insulation resistance between each of the following listed terminals and all others listed and case shall be greater than 20 megohms when tested as specified in 4.4.2.2. Unused terminals may be connected together and tested as one terminal.

P1- A, C, F, G, H, K, L, M, T, U, V, Y, Z, a, e, f, g, h, i, j, k, m, n, p, q, r, s, u, v, w, x, y, z, AA, BB, EE, FF, GG, HH, JJ, KK, LL, MM, NN, and PP.

Unused terminals: B, D, E, J, N, P, R, S, X, CC, b, c, d, and t.

J1- H, P, K, AA, BB, CC and DD.

Unused terminals: A, B, E, F, J, S, T, U, V, W and X.

J2- E, H, J, K, L, M, N, S, T, U and V.
Unused terminals: C, D and F.

J3- H, J, K, L, M, N, R, S, T, U and V
Unused terminals: C, D, E, and f.

J4- Unused terminals; B, D, F, L, M and N.

J5- T; Unused terminals: S, V, X, Y, Z, BB, CC and DD.

J6- S; Unused terminals: C, D, K, L, M. N, P, X, AA and BB.

J7- S, Y, V, U and Z.
Unused terminals: H, J, K, L, M and N.

J8- Unused terminals: B, D, E, F, H, J, K and M.

4.3.2.1 Classification of characteristics.
4.3.2.1.1 Rack and panel assembly (6A2A9) (see dwg 8833326).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect location or orientation of module receptacle connectors with respect to alignment pin holes	Gage
	102. Incorrect location or orientation of external connector	Visual

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103. Bent, loose or damaged connectors...Visual
104. Defective solder joints.....Visual/Manual
105. Component missing.....Visual
106. Loose or missing fasteners.....Visual/Manual
107. Cracked or broken connectors.....Visual
108. Board mounting screws loose.....Manual
109. Objective evidence relative to material improper or missing (see 3.1).....Visual
110. Serialization improper or missing (see 3.4).....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual
202. Identification markings illegible or obliterated.....Visual
203. Improper finish or color.....Visual

4.3.2.1.2 Preparation for delivery (see dwg 8858809).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Packing material and packaging not proper or missing	Visual
	102. Marking not in accordance with dwg	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.
4.4.1 Calibration capability. Not applicable.
4.4.2 Operational limits. With the specified voltages applied, perform the functional checks as specified in 3.5.2. Continuity shall be determined by a suitable method.
4.4.2.1 Continuity and resistance. Resistance measurements, indicating continuity between the terminals listed in Table I of 3.5.3, shall be measured using a Keithley Model 502 ohmmeter or equivalent.
4.4.2.2 Insulation resistance. The insulation resistance between the terminals listed in 3.5.4 shall be measured using a test voltage of 500 + 50 volts dc. Current shall be limited to 5 ma for this test.

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Table I

Primary Terminals	Secondary Terminals	Resistance (ohms)
J1-C	P1-V	0.5 max
J1-L	P1-s	0.5 max
J1-Y	P1-n	0.5 max
J1-AA	J3-B	0.5 max
J1-CC	J2-B	0.5 max
J1-BB	J3-A	0.5 max
J1-DD	J2-A	0.5 max
J1-Z	P1-m	0.5 max
J2-U	J5-A	0.5 max
J2-V	J5-B	0.5 max
J2-S	J5-C	0.5 max
J2-T	J5-D	0.5 max
J2-R	J5-E	0.5 max
J2-P	J5-F	0.5 max
J2-N	J5-H	0.5 max
J2-M	J5-J	0.5 max
J2-K	J5-K	0.5 max
J2-L	J5-M	0.5 max
J2-J	J5-L	0.5 max
J2-H	J4-C	0.5 max
J2-E	J4-A	6.5 ± 0.5
J3-U	J6-A	0.5 max
J3-V	J6-B	27 ± 4
J3-S	J7-A	0.5 max
J3-T	J7-C	0.5 max
J3-R	J7-B	0.5 max
J3-P	J7-D	0.5 max
J3-N	J6-E	0.5 max
J3-M	J6-F	8.0 ± 1
J3-K	J6-J	0.5 max
J3-L	J6-H	1.3 ± 0.5
J3-J	J7-E	0.5 max
J3-H	J7-F	1.3 ± 0.5
J4-K	J7-V	0.5 max
J4-H	J5-T	0.5 max
J4-J	P1-v	0.5 max
J4-R	P1-w	0.5 max
J4-E	P1-a	0.5 max
J4-P	P1-A	0.5 max
J5-MM	P1-f	0.5 max
J5-NN	P1-x	0.5 max
J1-M	P1-r	0.5 max

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Primary Terminals	Secondary Terminals	Resistance (ohms)
J5-LL	P1-y	0.5 max
J5-JJ	P1-G	0.5 max
J5-FF	P1-H	0.5 max
J5-HH	P1-h	0.5 max
J5-EE	P1-PP	0.5 max
J5-KK	P1-KK	0.5 max
J5-AA	P1-JJ	0.5 max
J5-W	P1-HH	0.5 max
J5-P	J7-Z	0.5 max
J5-U	P1-Z	0.5 max
J5-R	J7-U	0.5 max
J5-N	P1-Y	0.5 max
J6-CC	P1-C	0.5 max
J6-DD	P1-e	0.5 max
J6-Y	P1-AA	0.5 max
J6-W	P1-BB	0.5 max
J6-Z	P1-z	0.5 max
J6-X	P1-LL	0.5 max
J6-V	P1-j	0.5 max
J6-T	P1-k	0.5 max
J6-U	P1-q	0.5 max
J6-R	P1-F	0.5 max
J7-Y	J8-C	0.5 max
J7-X	P1-MM	0.5 max
J7-W	P1-NN	0.5 max
J7-S	J8-A	0.5 max
J7-R	P1-u	0.5 max
J7-P	P1-GG	0.5 max
J7-AA	P1-K	0.5 max
J7-CC	P1-i	0.5 max
J7-BB	P1-M	0.5 max
J7-DD	P1-L	0.5 max
J8-R	P1-U	0.5 max
J8-P	P1-q	0.5 max
J8-N	P1-T	0.5 max
J8-L	P1-p	1.3 ± 0.5

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